

PROJECT PROPOSAL

PHASE 2



E-RANK: DIGITAL TAXI RANK MANAGEMENT AND PASSENGER SAFETY SYSTEM

GROUP MEMBERS:

Oarabetse Morata 202406427
Phuti Setati 202435062
Kholofelo Phalakatsela 202306829
Louisa Mdluli 202324412
Siyabonga Ndzobondzobo 202441850

Group Name: PMP Solutions

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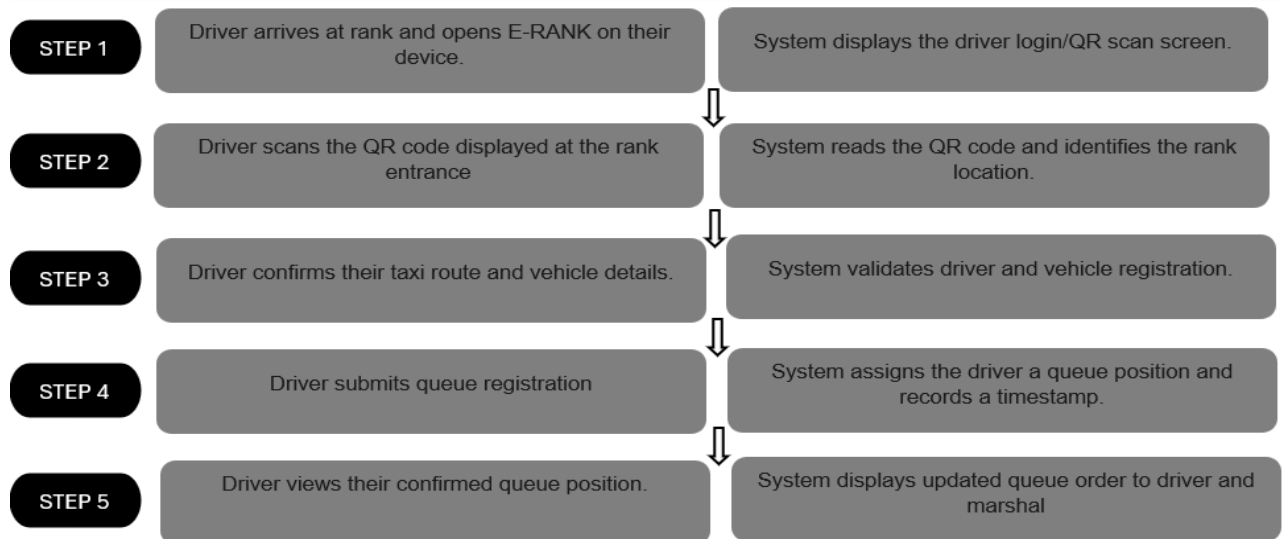
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1.Business Processes

This section describes the key operational processes within the E-RANK system and maps each process to the specific tasks performed by the relevant users (actors). Each process follows an activity flow that identifies who does what and when.

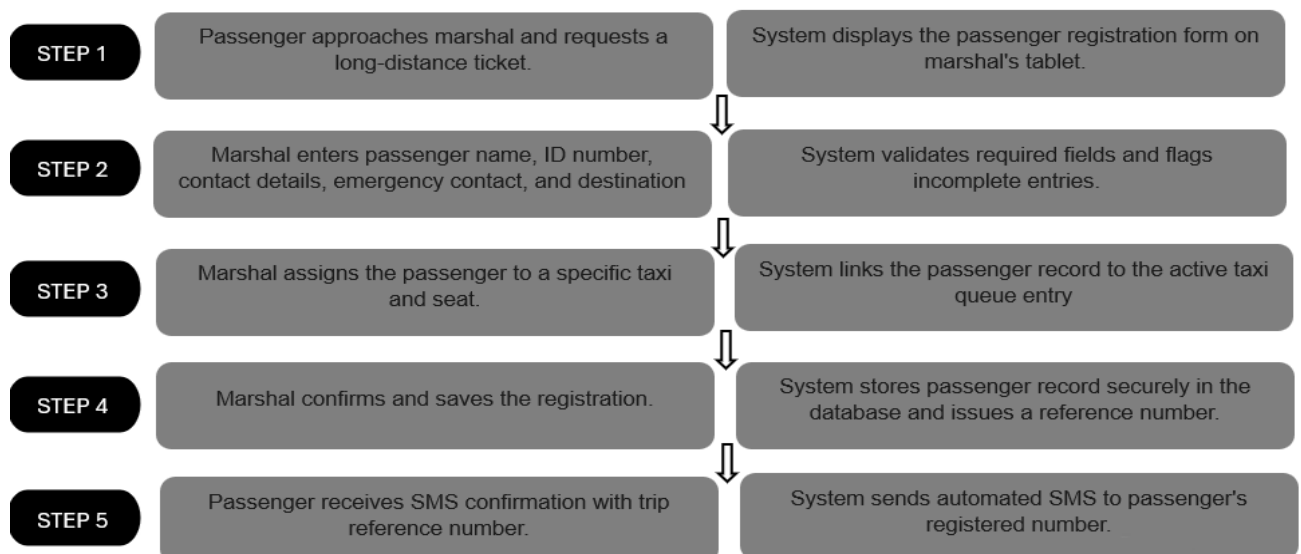
1.1 Driver Queue Registration Process

This process handles how drivers register their position in the digital taxi queue upon arriving at the rank.



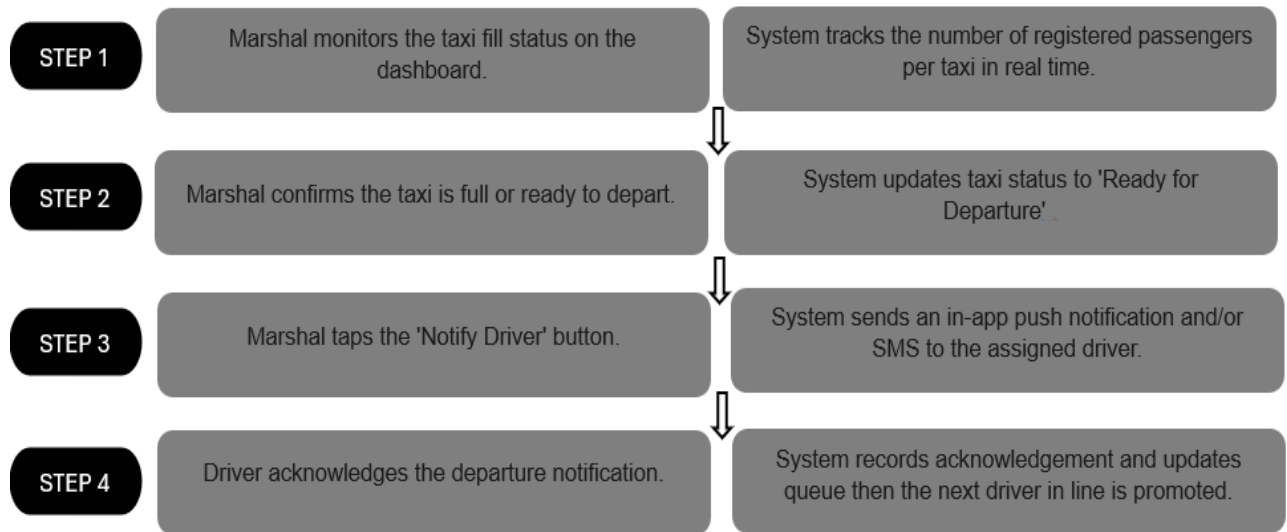
1.2 Passenger Registration for Long-Distance Travel

This process covers how the taxi marshal registers passenger details for long-distance trips, replacing the physical logbook.



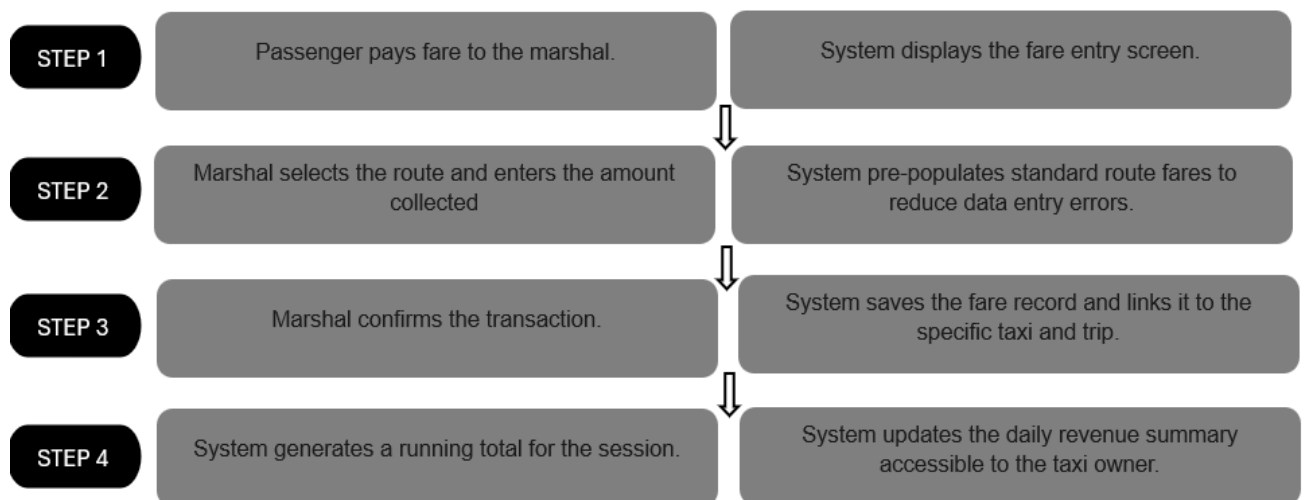
1.3 Taxi Departure Notification Process

This process describes how the marshal notifies a driver when their taxi is full and ready to depart.



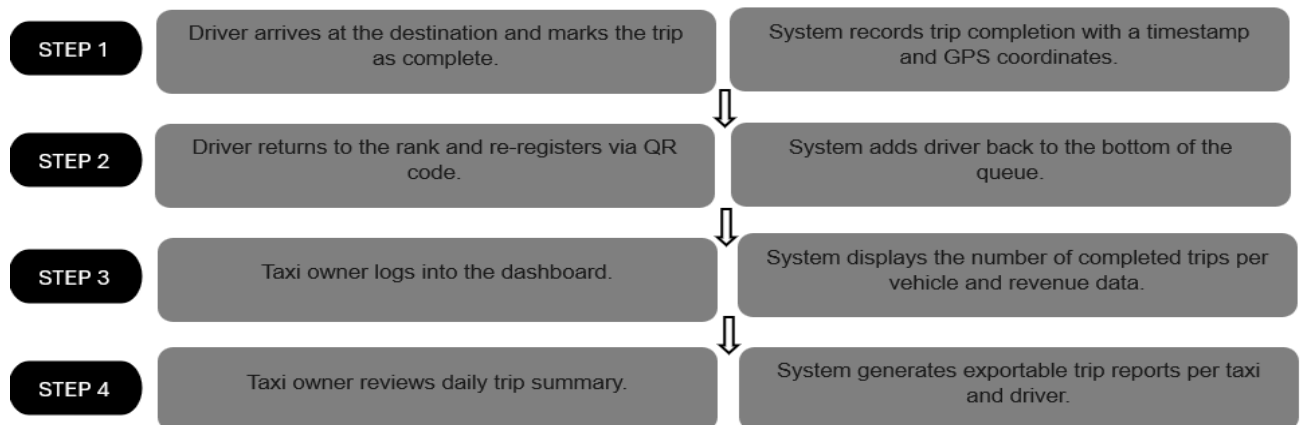
1.4 Fare Collection Recording Process

This process tracks how the marshal records fares collected from passengers at the rank.



1.5 Trip Confirmation and Monitoring Process

This process describes how a driver confirms a completed trip and how taxi owners monitor activity.



2. Non-functional requirements

Non-functional requirements define the quality attributes that the E-RANK system must exhibit. These requirements govern how the system performs its functions rather than what functions it performs.

2.1 Reliability

Reliability refers to the ability of the system to function correctly and consistently under expected operating conditions.

- ❖ The system must remain operational during peak taxi rank hours (05:00–20:00).
 - 99% uptime during operating hours.
- ❖ The system must not lose passenger registration data in the event of a network interruption.
 - Data is stored locally and synced when connectivity is restored.
- ❖ The system must recover automatically from minor crashes within 30 seconds.
 - Crash recovery tested across 50 simulated failures.
- ❖ The queue management module must maintain accurate ordering even during concurrent driver registrations.
 - No queue ordering errors during load testing with 20+ simultaneous users

2.2 Availability

Availability defines how accessible the system is to its users at any given time, including outside of peak hours.

- ❖ The system must be available 24/7 with planned maintenance windows scheduled outside operating hours.
 - Maximum 2 hours of planned downtime per week, outside 05:00–20:00.
- ❖ The system must support offline functionality for essential operations such as passenger registration when internet connectivity is unavailable.
 - Core functions accessible without live internet connection.

- ❖ The system must provide access to passenger records and taxi queues even during partial server outages.
 - Local cache maintains read access to last-synced data.

2.3 Security

Security requirements define how the system protects sensitive user data and prevents unauthorised access to system functions.

- ❖ All passenger personal data must be encrypted in transit and at rest using AES-256 encryption.
 - Verified by security audit and penetration test.
- ❖ The system must authenticate all users (marshals, drivers, owners) using unique credentials before granting access.
 - No unauthenticated access to any system module.
- ❖ Role-based access control must be enforced so that each user type can only access functions relevant to their role.
 - Marshals cannot access owner dashboards; drivers cannot view other passengers' details.
- ❖ The system must log all user actions with timestamps for audit purposes.
 - Audit log accessible only to system administrator.
- ❖ SMS notifications must be sent to verified phone numbers only.
 - Phone number verified via OTP during account creation.

2.4 Maintainability

Maintainability defines how easily the system can be updated, corrected, or extended by developers and system administrators.

- ❖ The system must be built using a modular architecture so that individual components (queue, passenger registration, notifications) can be updated independently.
 - Each module deployable without full system restart.
- ❖ All source code must be documented with inline comments and accompanied by a developer guide.
 - Documentation reviewed and approved before release.
- ❖ The system must support bug fixes deployable within 4 hours of an identified critical defect.
 - Critical patch deployment SLA of 4 hours.
- ❖ The system must include automated unit and integration tests with at least 80% code coverage.
 - Test coverage report generated per release cycle.

2.5 Portability

Portability defines the ability of the system to operate across different devices, operating systems, and network environments typical of South African taxi ranks.

- ❖ The marshal and owner interfaces must be accessible via standard web browsers on tablets and desktops (Chrome, Firefox, Edge).
 - Tested and verified on latest versions of all three browsers.
- ❖ The driver-facing QR scan functionality must work on Android devices running Android 9.0 and above.
 - Tested on a minimum of five different Android device models.

- ❖ The system must function on both 3G and 4G mobile data networks.
 - Functional testing conducted on 3G-equivalent bandwidth.
- ❖ The system database and backend must be deployable to major cloud providers (AWS, Google Cloud, Azure) without platform-specific code changes.
 - Cloud-agnostic deployment verified on at least two providers.

2.6 Performance

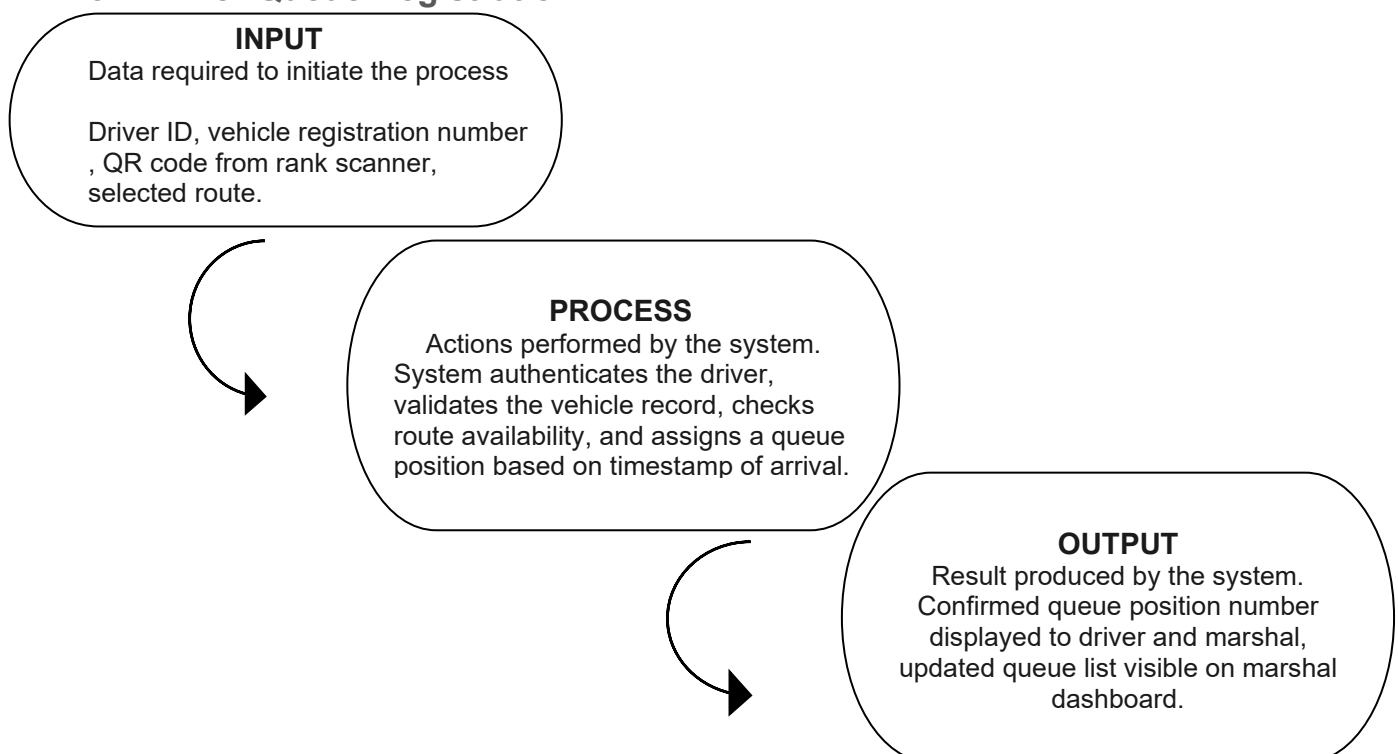
Performance requirements define the speed, responsiveness, and capacity standards that the E-RANK system must meet during normal and peak operation.

- ❖ The system must load any screen within 3 seconds on a standard 4G mobile connection.
 - Verified via load time testing across 100 sessions.
- ❖ The queue registration process must complete within 5 seconds from QR scan to confirmed position.
 - End-to-end transaction time measured and logged.
- ❖ The system must support at least 50 concurrent users per taxi rank without performance degradation.
 - Load tested at 50 simultaneous sessions with no response time increase beyond 10%.
- ❖ SMS departure notifications must be delivered within 30 seconds of the marshal triggering the alert.
 - Delivery timestamps logged and verified.
- ❖ The owner dashboard must generate a daily trip report within 10 seconds of request.
 - Report generation tested across datasets of up to 200 trip records.

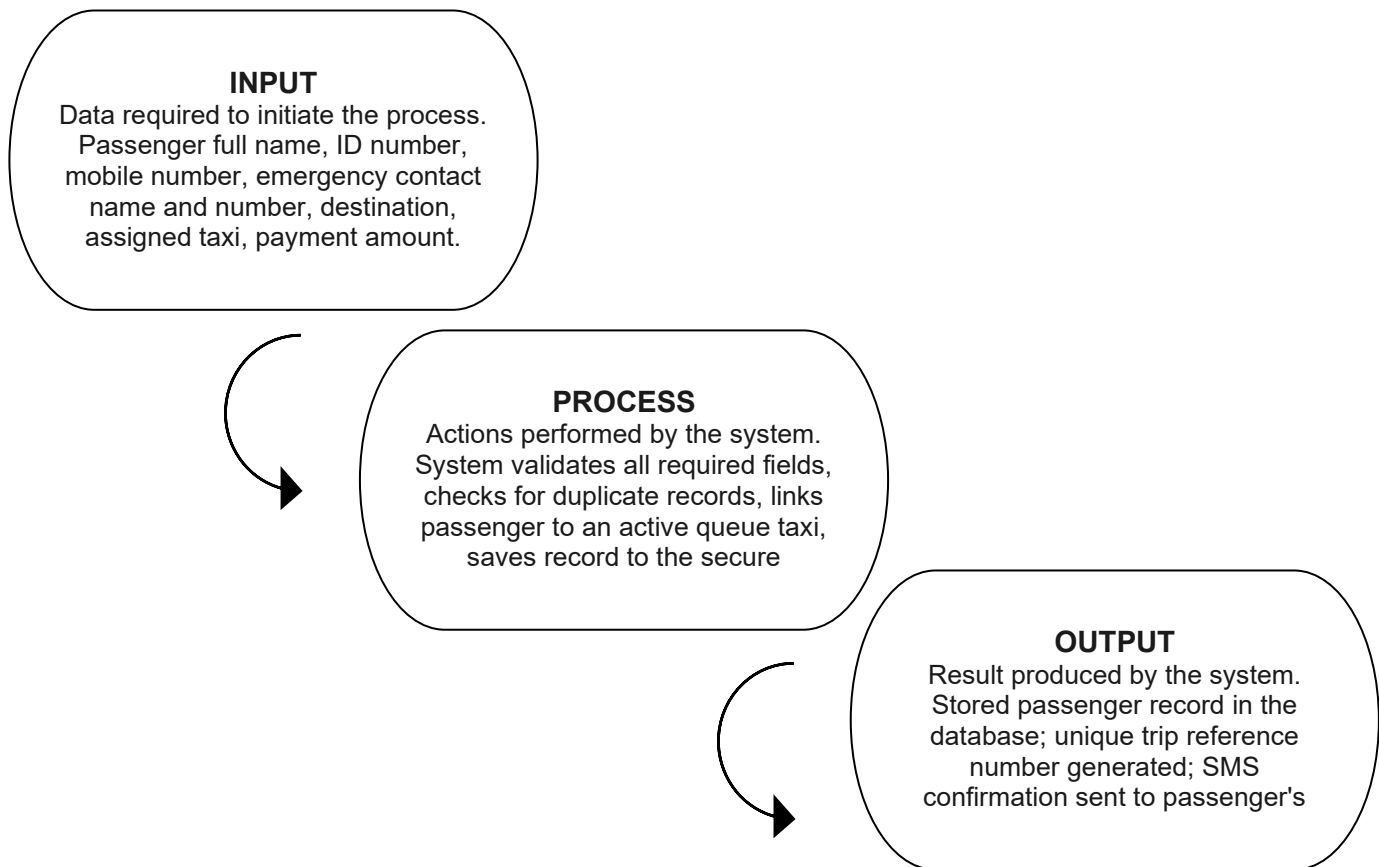
3. Functional requirements

Functional requirements specify the specific inputs, processes, and outputs that the E-RANK system must support. Each requirement is presented using the Input – Process – Output (IPO) model to clearly describe data flow through the system.

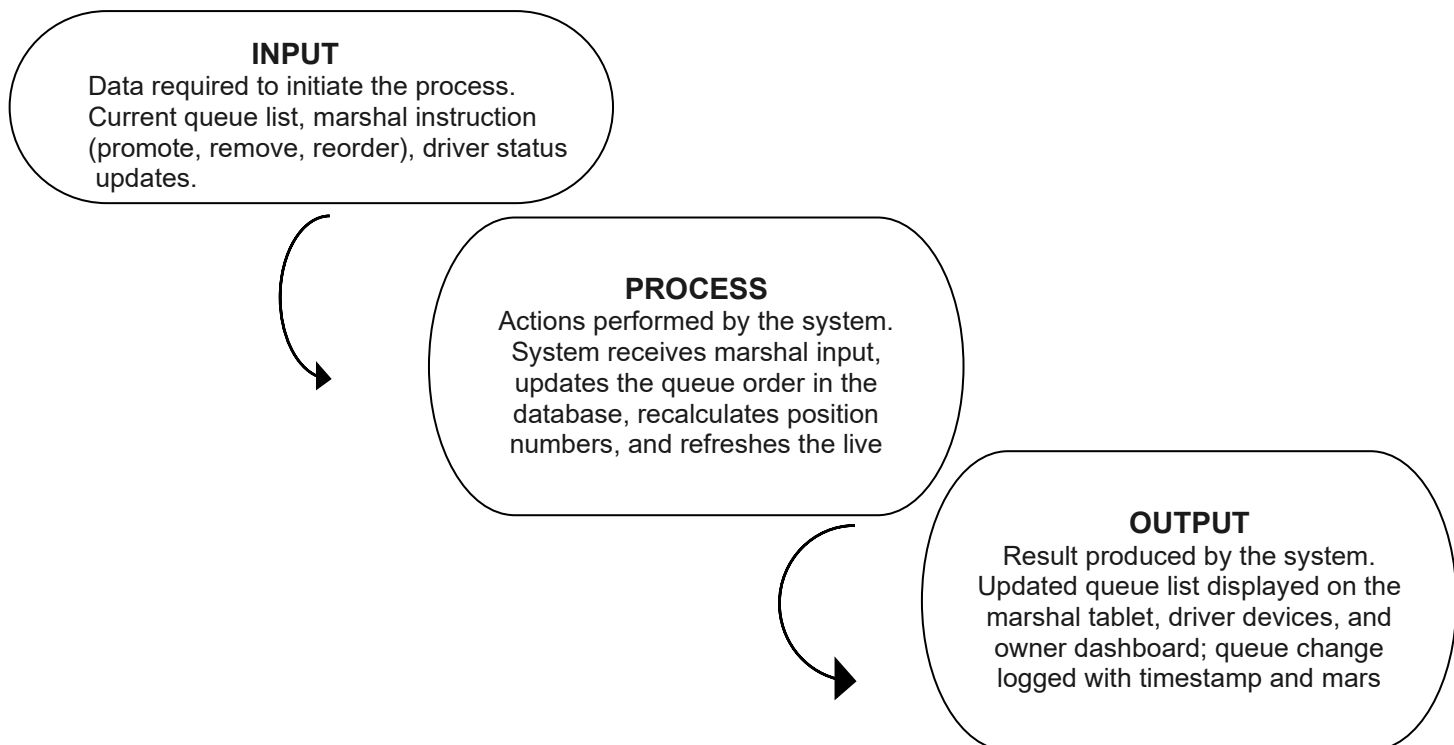
3.1 Driver Queue Registration



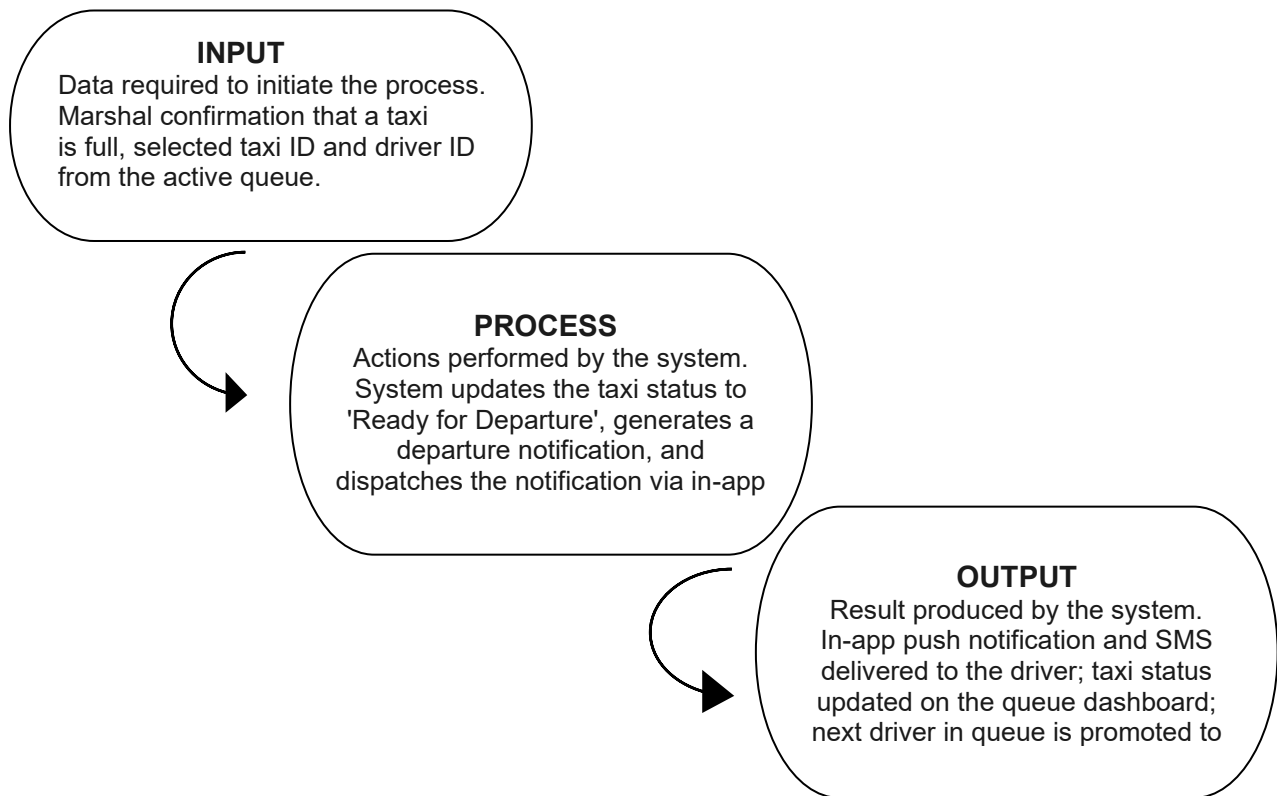
3.2 Passenger Registration for Long-Distance Travel



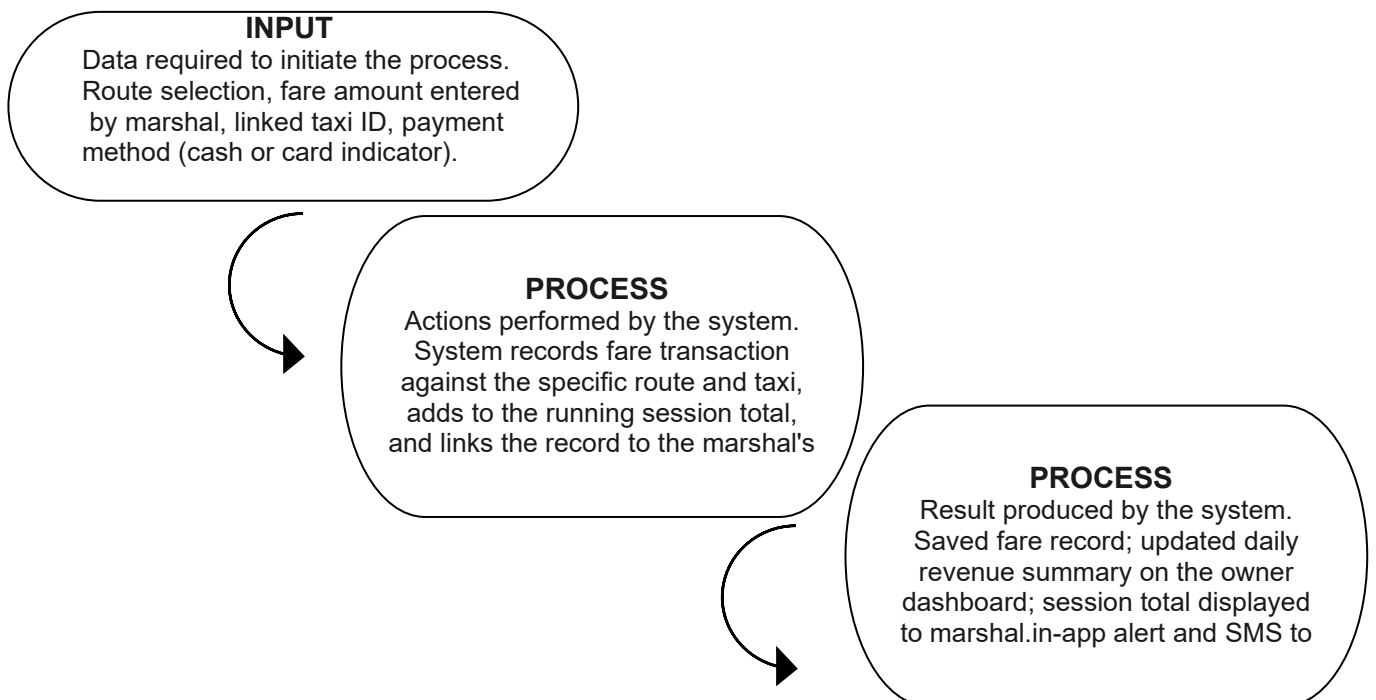
3.3 Queue Management



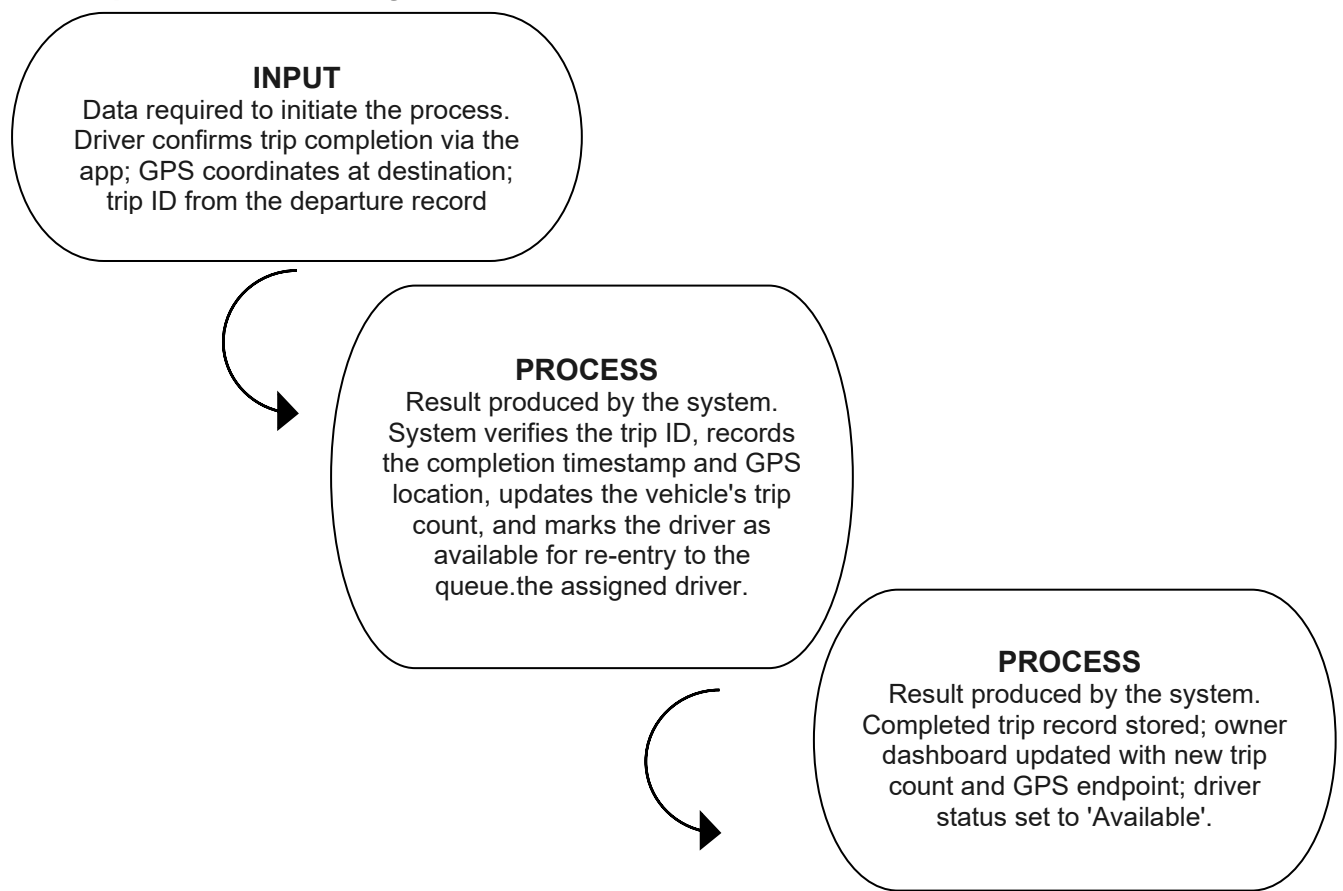
3.4 Departure Notification



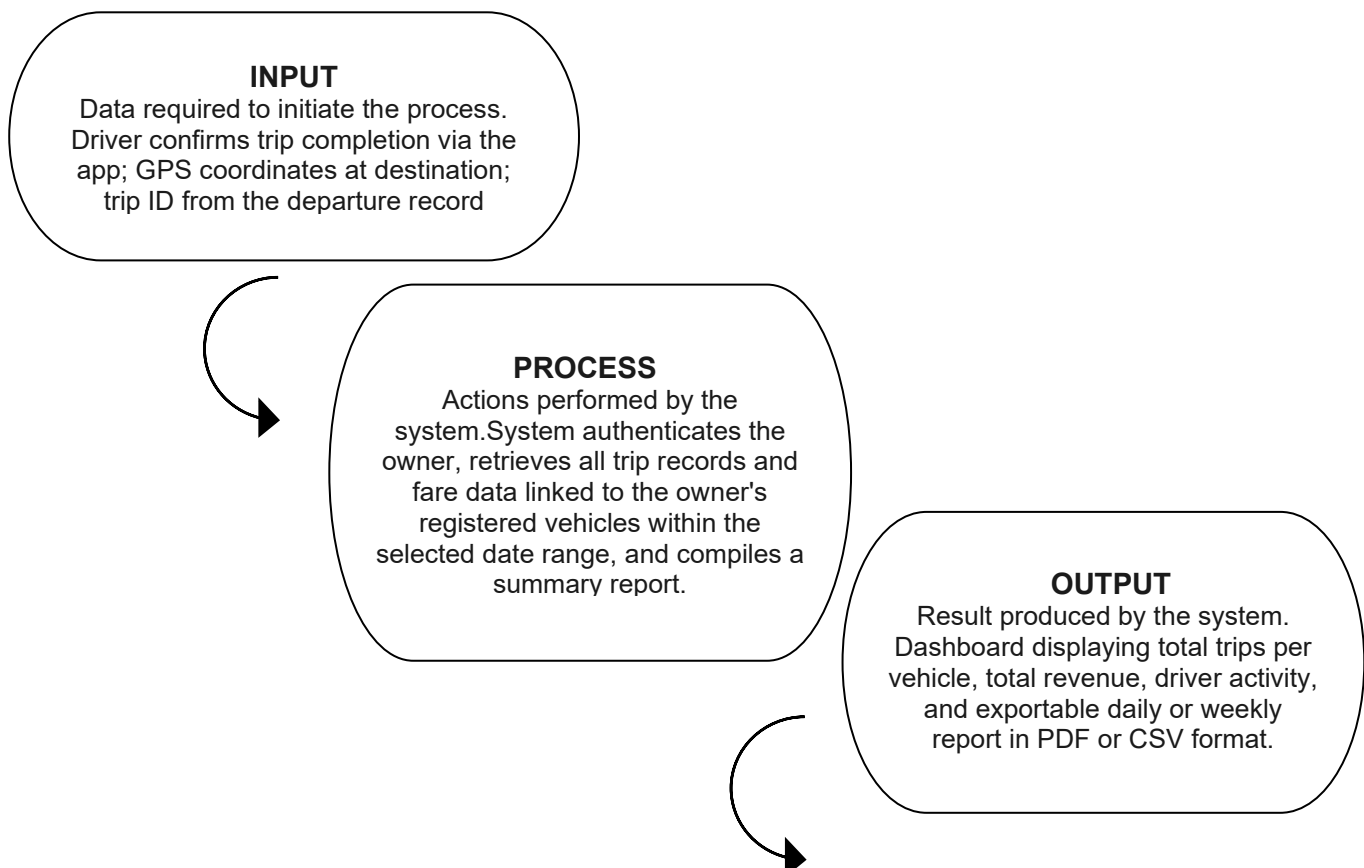
3.5 Fare Collection Recording



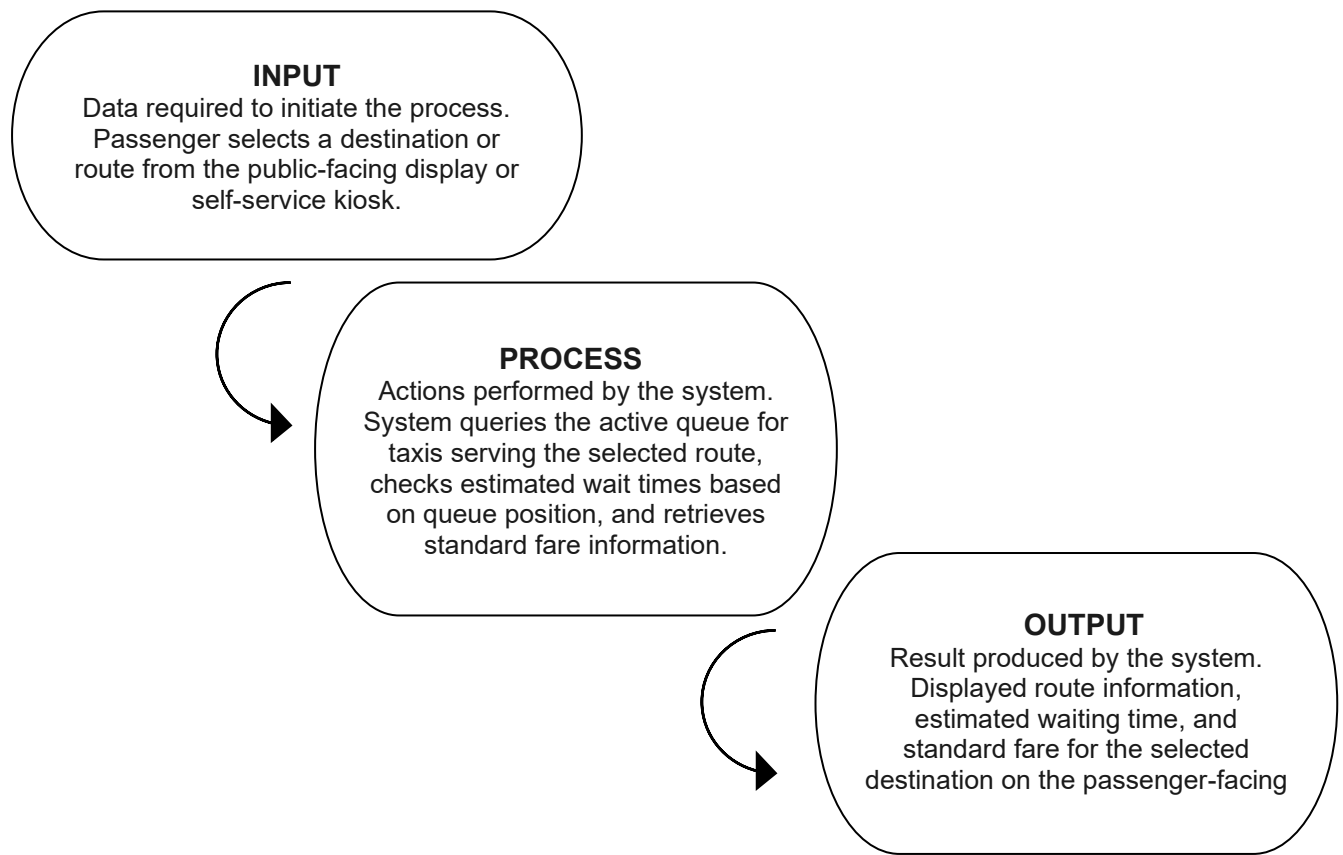
3.6 Trip Confirmation by Driver



3.7 Owner Trip Monitoring Dashboard



3.8 Route and Availability Information for Passengers



4. UML Diagrams

Figure 4.1: Use Case Diagram showing all key users:

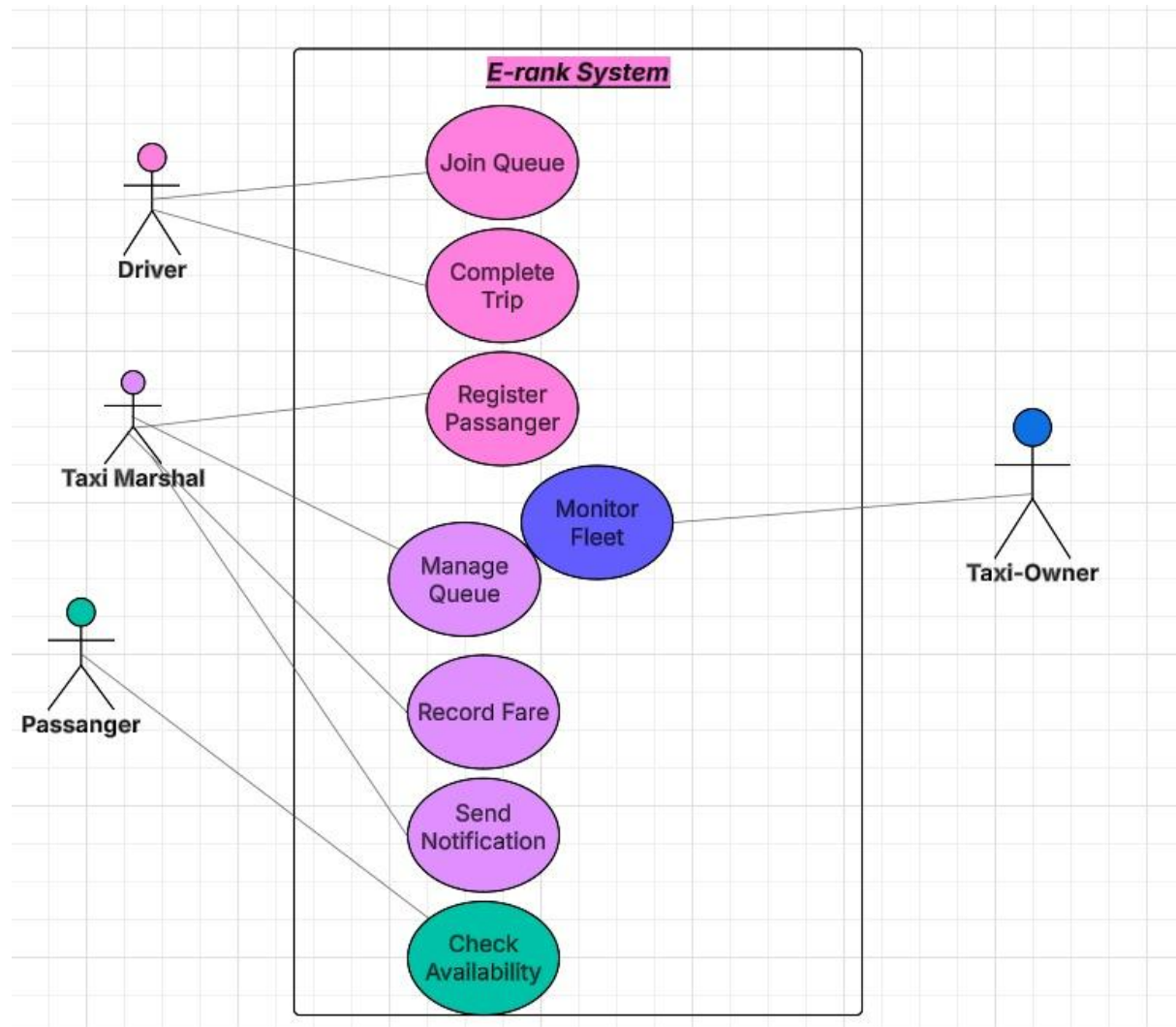


Figure 4.2: Context Diagram (System Boundaries):

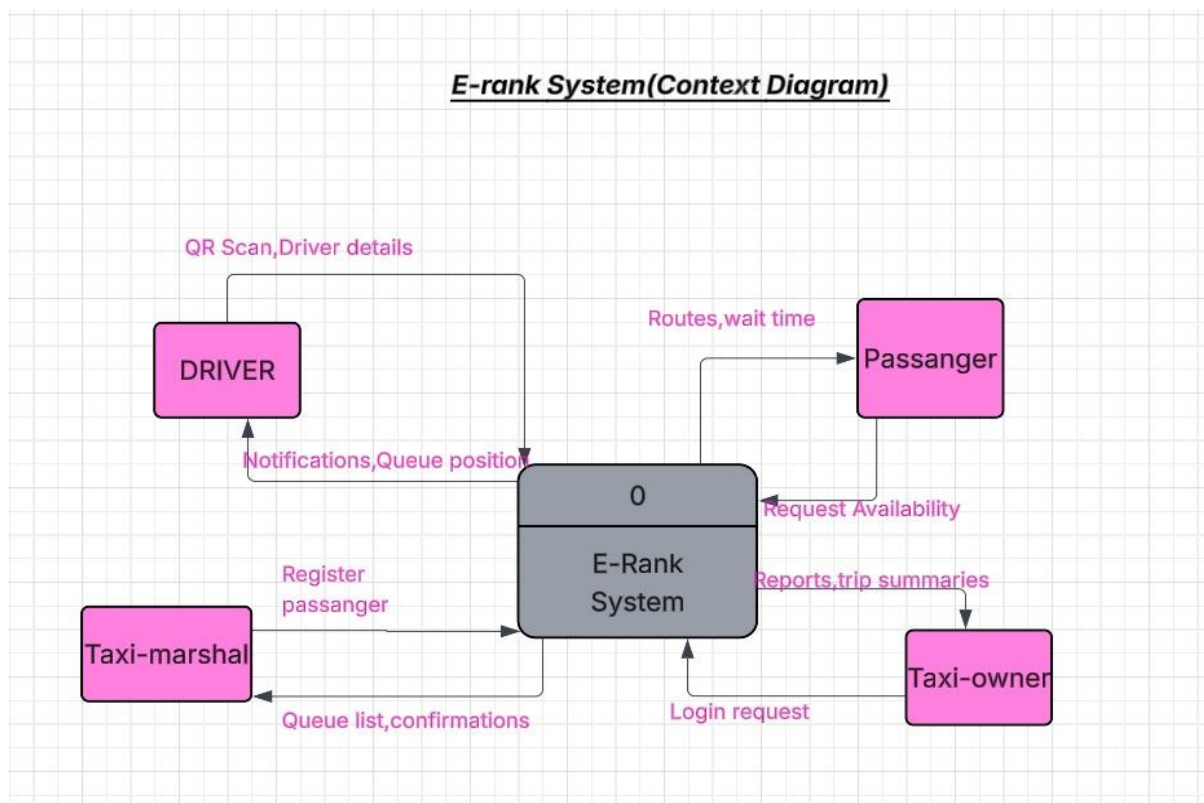


Figure 4.3: State Diagram – Driver State:

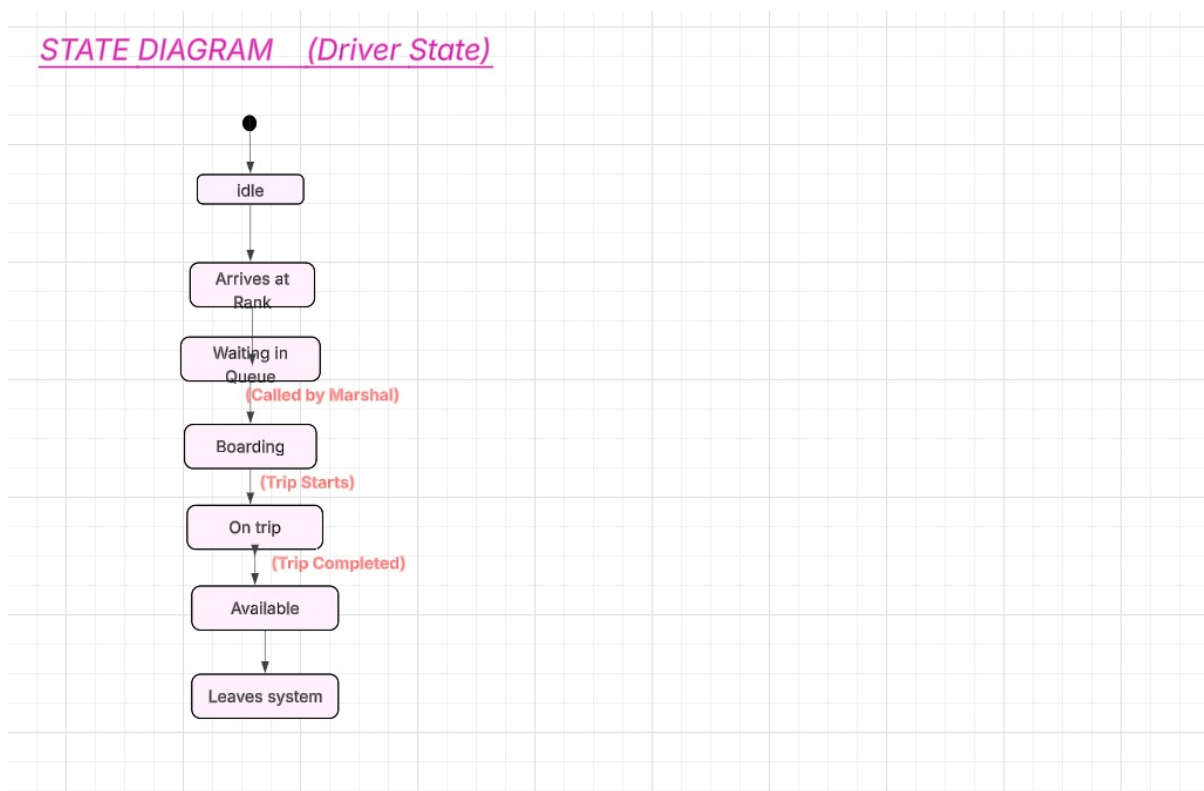


Figure 4.4: Sequence Diagram – Passenger Registration:

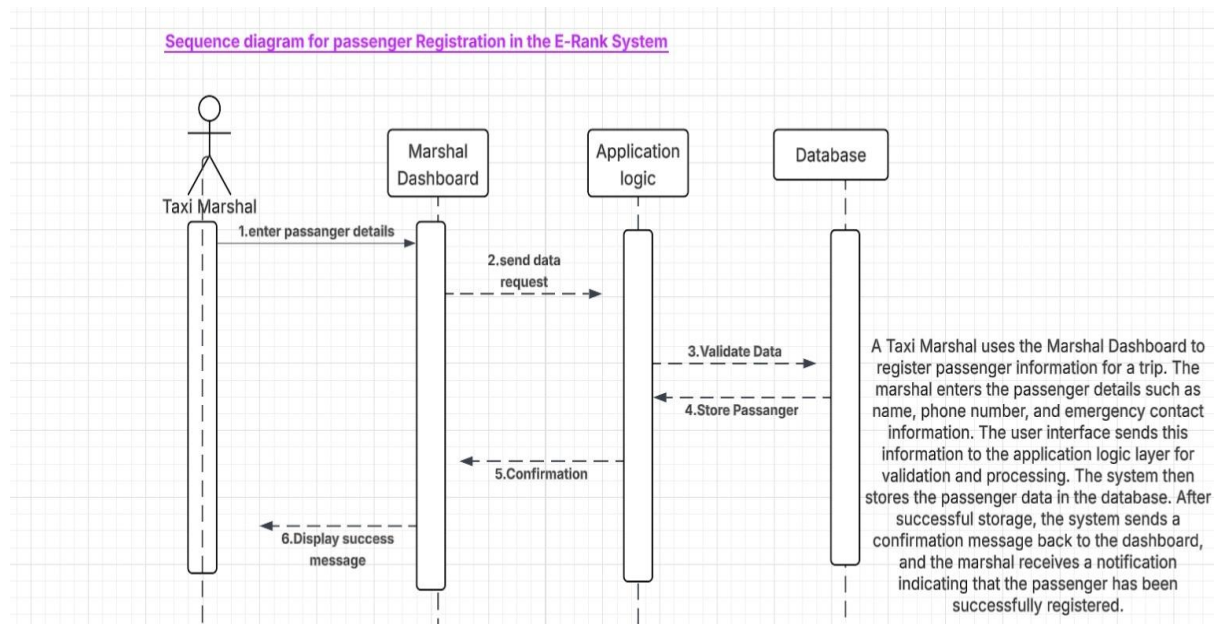
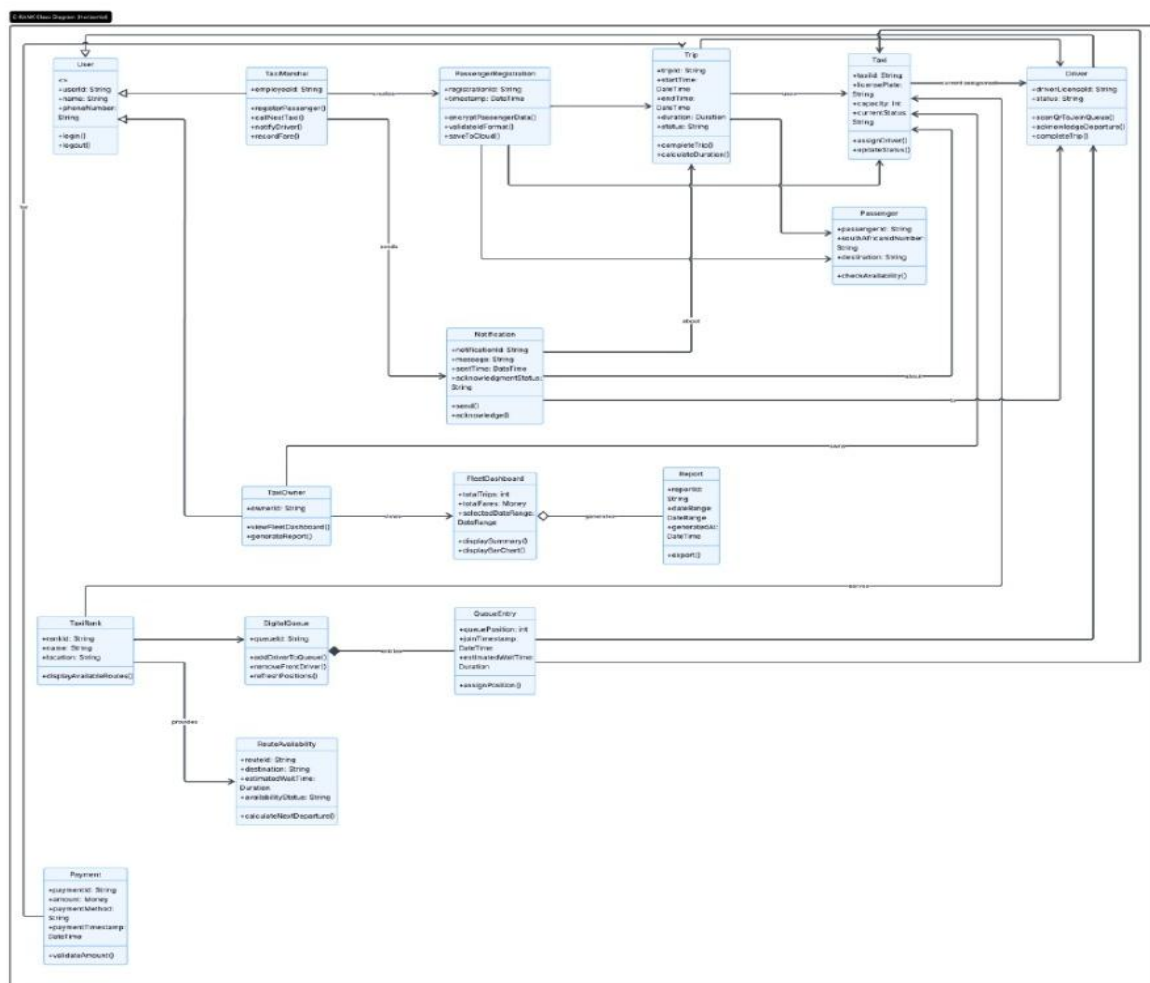


Figure 4.5: Class Diagram:



5. Project Management Artefacts

This section presents the key project management artefacts developed to ensure successful planning, execution, and monitoring of the E-RANK system. The Work Breakdown Structure (WBS) provides a detailed hierarchical decomposition of all tasks, deliverables, milestones, resources, dependencies, and risks. The Gantt Chart offers a clear visual timeline of the entire project schedule across the four phases.

Figure 6.1: Work Breakdown Structure:

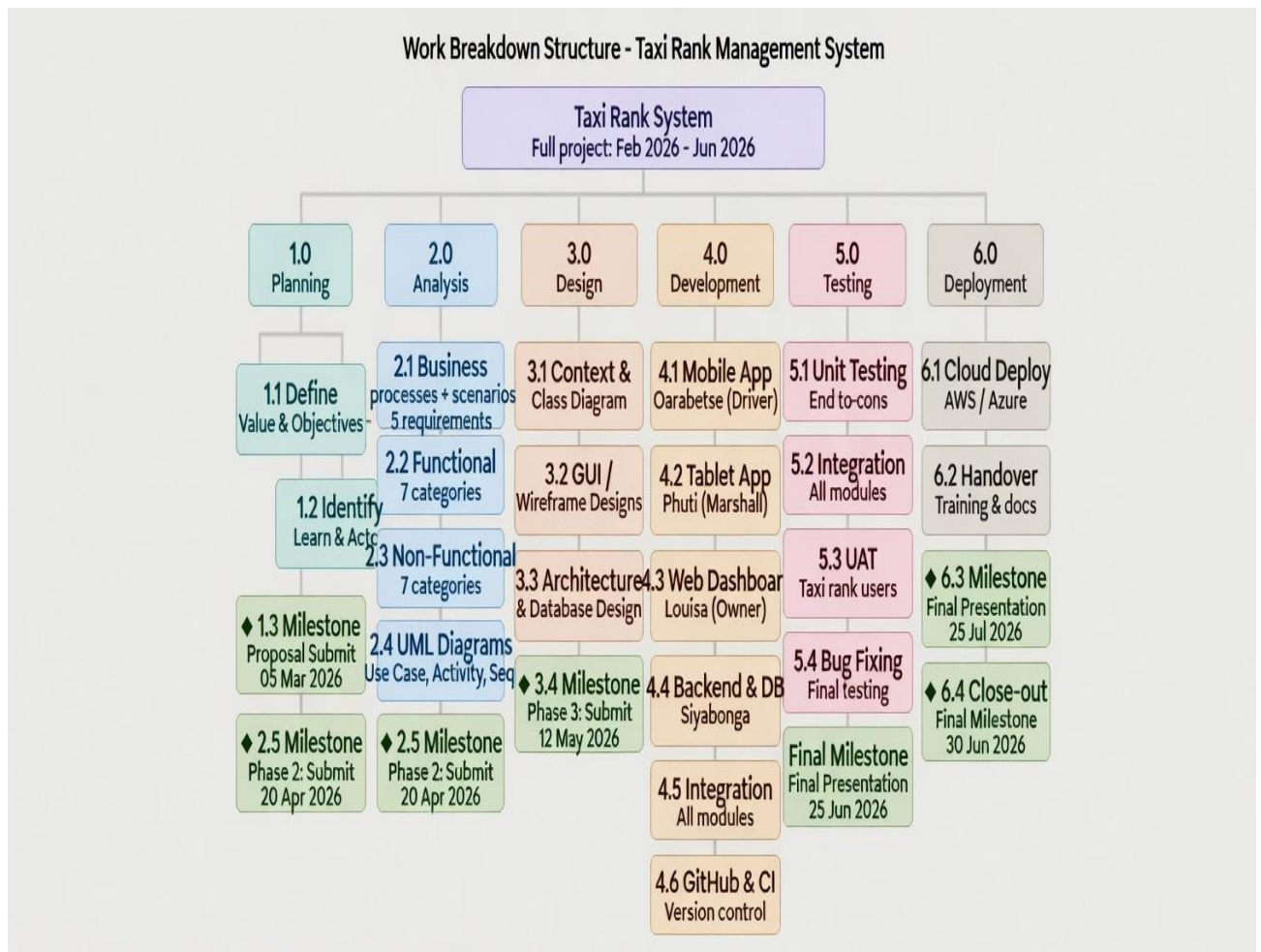


Figure 6.2: Gantt Chart – Project Timeline:

